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Sustained Attention Task (gradCPT) Dataset using simultaneous EEG-fMRI and DTI

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To study human attentional fluctuations, this study introduces Sustained Attention Task (the gradual onset continuous performance: gradCPT) multimodal dataset combining electroencephalography (EEG), functional magnetic resonance imaging (fMRI), and diffusion-weighted imaging (DWI). The dataset contains neuroimaging data from 28 participants across the attentional tasks (gradCPT, gradCPT with imagery), imagery task, visual task (flickering checkerboard), and resting-states (eyes-open and eyes-closed). We publicly share raw and preprocessed data from each modality to expand the scope of exploring the brain states during attentional fluctuations in the human brain. The accessibility of this dataset will provide opportunities for future research in investigating the relationship between attention dynamics and brain activity across different imaging modalities.

Background & Summary

Simultaneous acquisition of EEG and fMRI is advantageous for studying the function and states of the human brain. fMRI provides high spatial resolution (~mm)¹ but is limited by slow temporal dynamics (~s). In contrast, EEG offers high temporal resolution (~ms) but its spatial resolution is restricted (~cm) due to volume conduction via head tissues². While integrating these two techniques with complementary characteristics provides a more precise understanding of brain dynamics, the researchers must control a series of technical challenges.

From the EEG perspective, collecting EEG in the MR scanner environment can risk being exposed to various sources of noise which may greatly impact the data signal. The primary noise source comes from the MR gradient currents during the acquisition of fMRI data, whereby currents can be induced into the EEG electrodes—called gradient artifact. Another noise source arises from the ballistocardiogram (BCG) signal where the movement of EEG electrodes is caused by the arteries pulsation in the scalp which generates voltage. The BCG artifact increases in the strong magnetic field and with the strength of the field³. Besides the artifacts from the gradient and BCG, other sources of noise arise from the movement of the head, respiratory artifacts, and blinks of the eyes. From the fMRI perspective, the main challenge is the distortion of BOLD signals and degradation of the MR image quality induced by the existence of EEG materials⁴. To overcome these technical challenges of the simultaneous EEG and fMRI and secure data quality, this paper adapts the advanced processing tools based on the integration strategies presented by Telesford *et al.*⁵.

In this paper, we introduce a dataset⁶ gathered at the Center for Neuroscience Imaging Research, N Center of Sungkyunkwan University in Suwon, South Korea, consisting of simultaneous EEG-fMRI and DTI recordings from healthy adults. This dataset is unique compared to other public EEG-fMRI datasets (e.g., Telesford *et al.*⁵) as it helps to track attentional fluctuations over time with both high temporal (EEG) and spatial (fMRI) resolution using the sustained attention task (the gradual onset continuous performance task, gradCPT). As gradCPT measures instantaneous distraction or reaction time changes during sustained attention, it is suitable for

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Recording Condition	Tasks	Run # (per Task)	Task Order	Run Length (s)	TR Length
Scan OFF	Eyes open	Run 1	*	180	—
	Eyes closed			200 (15 s + 35 s) × 4	
	Checkerboard (15 Hz)				
	gradCPT				
Scan ON	Checkerboard (15 Hz)	Run 1	*	350 (15 s + 35 s) × 7	183
	gradCPT	Run 1	*	360	
	Imagery	Run 1	*		
	gradCPT	Run 2	random		
	Eyes closed	Run 1	random		
	gradCPT with Imagery	Run 1	random		
	gradCPT	Run 3	random		
	Imagery	Run 2	random		
	gradCPT with Imagery	Run 2	random		
	Eyes open	Run 1	random		

Table 1. Example of scanning protocol for simultaneous EEG-fMRI experiment. Participants performed all tasks under both Scan OFF and Scan ON conditions. Run # (per Task) represents the number of runs per task. The astronaut symbol, ‘*’, of Task Order column represents fixed (static) order. ‘random’ represents counterbalanced task presentation order across subjects. ‘(15 s + 35 s) × 4 or 7’ indicates four (Scan OFF) or seven (Scan ON) repetitions of a 15-second rest period following a 35-second stimulus presentation. TR (repetition time) was fixed at 2 seconds for all fMRI acquisitions.

observing time-varying attention states and brain activity. Additionally, the paradigm is designed to explore the interaction between internal and external attention by combining gradCPT with an imagery task that stimulates mind-wandering, characterized by reduced engagement with the external task and increased internal cognitive activity. Since fMRI has excellent spatial resolution but limited in temporal resolution, and EEG has opposite characteristics, this dataset combining the two techniques provides a strong basis for analyzing the dynamic mechanisms of the brain related to attention fluctuations.

The dataset comprises multiple task conditions within a single scan, including gradCPT only task, gradCPT with imagery dual task, an imagery only task, a simple visual task (viewing a flickering checkerboard pattern), and resting-state conditions (eyes-open and eyes-closed). This design allows for within-subject comparison within the same participant. In addition, all participants participated in the EEG-only session by performing the gradCPT, checkerboard, and rest (see Table 1) while the scanner being silent (Scan OFF). This EEG-only session was used as a baseline to evaluate the impact of artifacts occurring in the fMRI environment and to verify the refining performance of the simultaneous measurement EEG data.

This dataset also contains diffusion-weighted imaging (DWI) data, enabling to build structural brain connectivity of white-matter connections⁷. By providing high-resolution EEG, fMRI, and DWI data in an integrated manner, this dataset will be a valuable resource for researchers interested in attention, internal cognitive states, multitasking, and spatiotemporal interactions of brain networks. We describe the data collection process, pre-processing procedure, and quality control indicators, and publicly share raw and preprocessed data and codes through OpenNeuro⁶ and GitHub.

Methods

Participants. 28 participants took part in a single session simultaneous EEG-fMRI experiment (16 females; mean age 27.34 ± 6.58 years old), of whom 23 were right-handed. Our sample size (N = 28) aligns with previous studies using fMRI to investigate attentional processes (e.g., Kondo *et al.*⁸, N = 29). The sample size is sufficient for exploring attentional dynamics. All participants reported no history of claustrophobia and had normal or corrected-to-normal vision. Research participants were recruited through a recruitment notice and a survey targeting students (undergraduate and postgraduate) from Sungkyunkwan University and voluntary applicants. At least one day before the experiment, participants visited the laboratory for approximately 30 minutes for the orientation regarding the experimental procedure, which also included measuring head size to determine the appropriate EEG cap size. Participants unable to visit the site had an online orientation instead. All experiment was conducted with the approval of Sungkyunkwan University’s Institutional Review Board (SKKU IRB 2022-12-013) and written informed consent was obtained from all participants prior to the experiment, including formal consent for participation and for the use of their anonymized data in publications and open-access sharing. Additionally, the short form of the Big Five Inventory (BFI) and demographic information were collected from the participants. The short form of the Big Five Inventory (BFI) has been translated into Korean and developed to be suitable for Koreans, with its reliability and validity verified⁹. Due to varying data quality and specific analysis requirements, different subsets of participants were included in each analysis (e.g., 26 for fMRI quality checks, 21 for gradCPT, and 19 for checkerboard analyses). Details for each analysis are reported in the respective sections.

EEG acquisition. EEG data were recorded through an MR-compatible BrainCap MR system with 64 channels by Brain Products at a sampling rate of 5000 Hz. We used two BrainAmp MR amplifiers (32 channels for each amplifier) and a PowerPack battery. EEG data were collected from 63 cortical channels (one ECG (no.32)

Modality	Equipment	Sampling Rate	Additional Information
EEG/ECG	Brain Product BrainCap MR with Multirodes	5000 Hz	64 channels • EEG: 61 cortical electrodes • ECG: 1 channel Placed on participant's back (channel 32)
fMRI	Siemens 3.0 T Magnetom Prisma	TR: 2000 ms	BOLD fMRI Parameters • Slices: 40 • FOV: 224 × 224 × 140 mm • Voxel Size: 3.5 mm ³
Stimulus Presentation	1920*1080 Propixx		

Table 2. Equipment used for data collection and stimuli presentation in EEG-fMRI study.

channel) using BrainVision Recorder (see Table 2). Female participants commonly used a 56 cm sized EEG cap and male participants used a 58 cm sized EEG cap. EEG channels were positioned according to the international 10-10 system—MR-compatible (EEG conductive) abrasive electrolyte-gel (ABRALYT 2000; Easycap GmbH, Herrschingen, Germany). Impedance values of the reference channel (FCz) and ground channel (AFz) were set to below 5k Ω and other connected electrodes were set to below 20 k Ω .

MRI data acquisition. fMRI image collection was acquired using a 3 T Siemens Magnetom Prisma equipped with a 64-channel head coil. MPRAGE T1-weighted (T1w) anatomical images were acquired with the following parameters: TR = 2400 ms; TI = 900 ms; echo time (TE) = 2.28 ms; flip angle = 8°; slices = 256; field of view (FOV) = 208 × 256 × 256 mm; voxel size = 1 mm³ isotropic; phase/slice partial Fourier off; pixel bandwidth = 200 Hz/Px. BOLD fMRI images were obtained with the following parameters: TR = 2000 ms; TE = 25 ms; flip angle = 73°; slices = 40; FOV = 224 × 224 × 140 mm; voxel size = 3.5 mm³. The fMRI monitor resolution was 1920 × 1080 (see Table 2).

DWI data acquisition. MRI scans for diffusion-weighted imaging (DWI) were collected with the following parameters: TR = 7000 ms; TE = 80 ms; slices = 64; FOV = 220 mm; voxel size = 2.5 mm³. Images were collected with 64-direction diffusion encoding ($b = 0$, 2500 s/mm²) and no diffusion encoding ($b = 0$ s/mm²). Most T1 and DWI scans were taken following the functional sequence, but depending on the participant's condition (i.e., drowsiness), these structural and DWI scans were also taken in between the tasks.

Simultaneous EEG-fMRI procedures. The scanning protocol included two recording conditions: “Scan OFF” and “Scan ON”. Both conditions were conducted inside the scanner whereby the scanning room was set to dim light. All participants first performed the tasks under the Scan OFF condition, followed by the Scan ON condition. In the Scan OFF condition, participants conducted the task with the MRI scanner being silent while the EEG data was being recorded. In the Scan ON condition, participants conducted the tasks during simultaneous EEG-fMRI recordings. The tasks for the Scan OFF condition were presented in a fixed order identically for all participants. In contrast, we partly counterbalanced the tasks in Scan ON condition, the first three tasks were presented in an identical order for all participants, while the remaining attentional tasks with multiple blocks were counterbalanced to reduce order effects, such as worse performance, and practice effects, to reduce the subject's enhancement of task performance due to familiarity. Resting-state conditions were performed between the tasks (see Table 1. for the data acquisition details). Eyelink 1000 was used to monitor participants' engagement in the task via eye tracking.

The EEG cap preparation time per participant was approximately 30–45 minutes. MR-compatible eyeglasses were provided to participants requiring vision correction before entering the scanner room. To minimize head motion inside the MRI scanner, participants were instructed to avoid moving their head and body once the scanning began. The total duration of the experiment did not exceed two hours in the fMRI room.

Two bundles of EEG cap cables were connected to the corresponding MR-compatible amplifiers, which were powered by a PowerPack battery. As recommended by the Brain Products, the two amplifiers and the battery were placed behind the participant's head¹⁰, aligned with the scanner's z-axis. Throughout the acquisition of the EEG data, EEG recording system is synchronized to the master clock of the MRI scanner. Data collection from EEG recording was with task onset triggers, and at the onset of each TR (every 4 seconds), volume triggers were recorded together with EEG signals for data synchronization (see Fig. 1). Sandbags were placed over the cables and amplifiers to minimize the vibrations.

Task data/stimuli description. This section describes the dataset acquired from the EEG-fMRI experiment, which includes resting-state, visual task, attentional task, and internal task data. All stimuli were created using Psychtoolbox^{11,12}, and executed in MATLAB (www.mathworks.com). The EEG and fMRI data were collected during simultaneous EEG recording and fMRI scanning (Scan ON condition), and EEG data were additionally acquired without the fMRI scanning to assess the influence of the fMRI scanning conditions on EEG signals (Scan OFF condition).

Resting-states. The resting-state condition consisted of eyes-closed and eyes-open sessions. These were performed both during the Scan OFF (EEG-only) and Scan ON (simultaneous EEG-fMRI). In the Scan OFF condition, the eyes-open session was always first performed, followed by eyes-closed session. However, in the Scan ON condition, the order of the two sessions was counterbalanced across participants to avoid potential order effects.

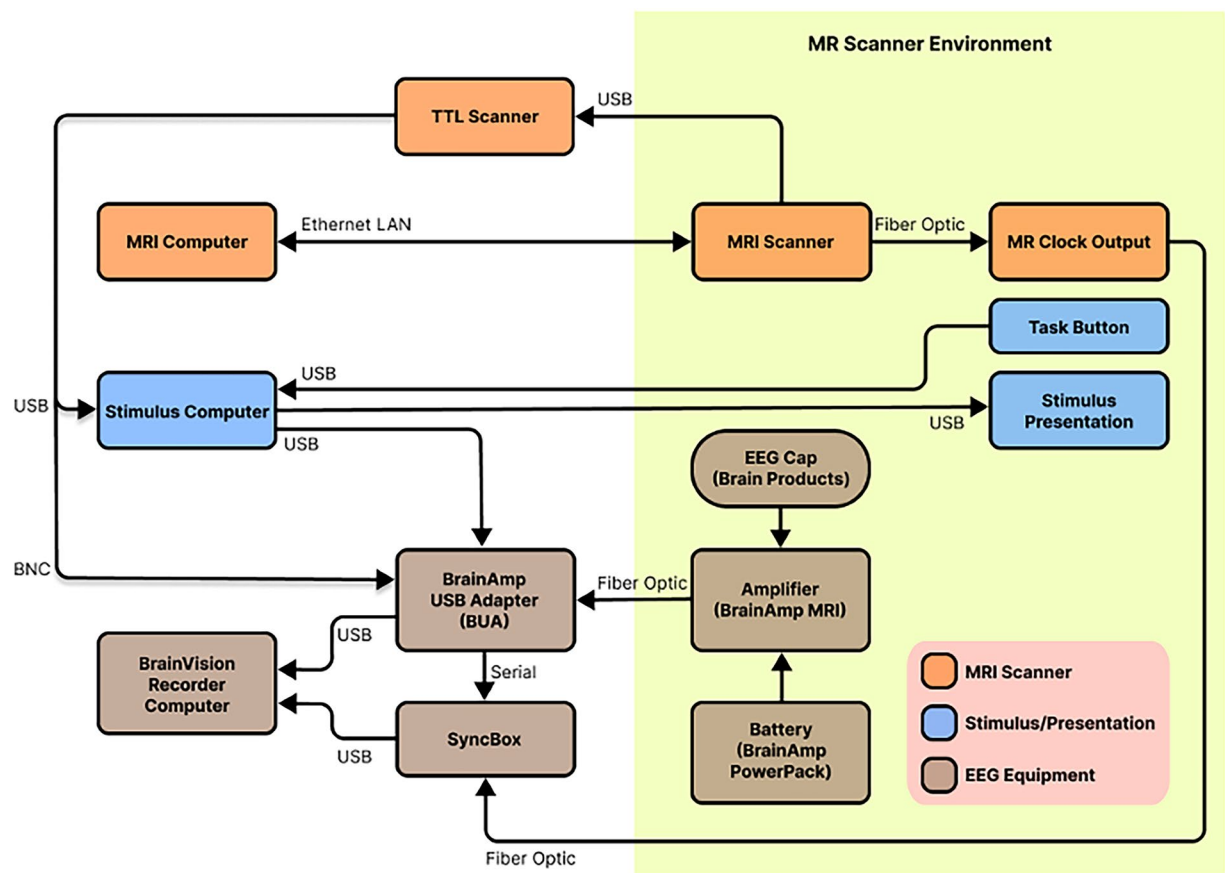


Fig. 1 Schematic of EEG-fMRI experiment setting/environment.

During the eyes-open session, participants viewed a white cross on a black background at the center of the screen and were instructed to keep their eyes open and maintain fixation on the white cross. Participants viewed the screen through a mirror mounted on the head coil. In the eyes-closed session, participants were guided to close their eyes, focus on their breathing, and be mindful not to fall asleep. Each resting-state session lasted 180-seconds in the Scan OFF conditions and 360-seconds in the Scan ON condition.

Visual task: Checkerboard stimulus. Checkerboard stimulus was developed by Telesford *et al.*⁵ Participants were presented with a high-spatial frequency of flickering circular checkerboard (black-and-white) at 15 Hertz under both task conditions. The flickering frequency was chosen to be separated from the alpha band (8–12 Hz). Each trial of the checkerboard stimulus lasted 35-seconds, followed by a 15-second rest period. The task was repeated 4 times during the Scan OFF condition and 7 times during the simultaneous EEG-fMRI (Scan ON) condition (see Fig. 2a). Participants were instructed to fixate on the checkerboard during its presentation and, during the subsequent rest period, to maintain fixation on a white cross positioned at the center of the screen formerly occupied by the checkerboard.

Code for presenting the checkerboard stimulus is available on GitHub. (https://github.com/NathanKlineInstitute/NATVIEW_EEGFMRI/blob/main/stimulus/natview_stimuli_checkerboard.m).

Attentional task: Sustained attention task (gradCPT). Originally developed by Esterman *et al.*¹³ as a paradigm for sustained attention, the version of the gradual onset continuous performance task (gradCPT) used in our experiment follows the refinement introduced by Jun *et al.*¹⁴. While Esterman *et al.*¹³ employed relatively small, low-resolution scene images which might have limited stimulus discriminability and potentially affected task performance, Jun and Lee (2021) improved the paradigm by using higher resolution and larger images. Code (by Jihyang Jun and Vanessa G. Lee) for presenting the gradCPT stimulus is available on GitHub (<https://github.com/MoonBrainLab/GradCPT-Simultaneous-EEG-fMRI-DTI-Data>). In this task, a sequence of scenes was gradually fade in and out. Each scene was displayed and gradually transitioned into the next scene in 800 ms, where the previous image fades out and the following image fades in (see Fig. 2b; and the accompanying movie in Esterman *et al.*¹³). The scenes were divided into two categories: city and mountain, with 10 grayscale photographs in each category. No scene was repeated on consecutive trials. Before the main experiment, participants were familiarized with the task by completing a 20-second practice session. During the main task, participants used a fiber-optic button box (Current Designs, Philadelphia, PA) to respond. They were required to press the button for city scenes, which appeared 75% of the time, and withhold responses for mountain scenes, which appeared 25%

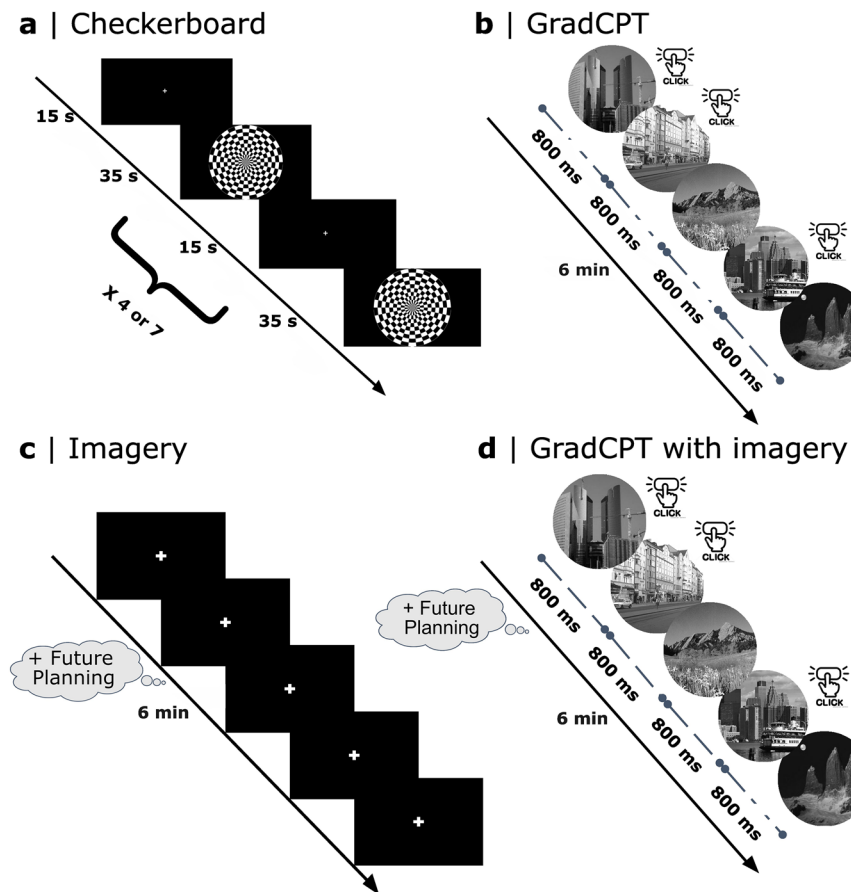


Fig. 2 Task paradigm. (a) checkerboard, (b) gradCPT (gradCPT only), (c) imagery, and (d) gradCPT with Imagery.

of the time. The task was repeated three times, with the presentation order counterbalanced to avoid consecutive repetitions. Each trial block lasted 360-seconds.

Imagery task. Participants performed a future-oriented internal cognitive task as in Cohen *et al.*¹⁵. Specifically, they were instructed to imagine research- or work-related activities they would perform at school or workplace on an upcoming day. They were encouraged to visualize their scenarios in as much details as possible while maintaining fixation on the crosshair (see Fig. 2c). At the 90th TR during the scan, participants were asked to describe what they were imagining at that moment, replying with a single word using a noise-cancelling microphone (FOMRI-III; Optoacoustics Ltd., Mazor, Israel) attached to the head coil. To ensure participants understood the task, they completed a 60-second practice session beforehand. The imagery task was repeated twice, with the presentation order counterbalanced to avoid consecutive repetition. Each block lasted 360-seconds.

Attentional task: Dual task (gradCPT with Imagery). In this condition, participants performed both the gradCPT and the imagery tasks simultaneously. While performing the gradCPT, they continuously imagined work- or school-related task they would complete on a specific upcoming day. Participants were instructed to focus on both tasks as best as they could. As in the imagery-only condition, the experimenter asked participants at the 90th TR to describe their current imagination in a single word. Sustained attention to internal cognition during the visual task was expected to reduce task performance, with measurable difference in behavior and brain network dynamics between “in-the-zone” and “out-of-the-zone” period¹³ (See Fig. 2d). This combined task was repeated twice, and the presentation order was counterbalanced to avoid consecutive repetition. Each trial block lasted 360-seconds.

EEG preprocessing. We collected EEG data in both Scan OFF and ON conditions inside the MRI scanner. The preprocessing method applied to the EEG data varied depending on whether the MRI was off or on. All EEG data were preprocessed with EEGLAB and its associated plugins. For the data, preprocessing was performed using the FMRIB plugin for EEGLAB¹⁶, provided by the University of Oxford’s Centre for Functional MRI of the Brain (FMRIB)¹⁷.

In the Scan OFF setting, the following preprocessing steps were used: Even though gradient artifacts of scanner on session were absent, we used artifact detection and removal techniques to remove signal contamination caused by the participant’s heartbeat¹⁸: (i) detection of QRS complexes/heartbeat through the ECG channel; (ii)

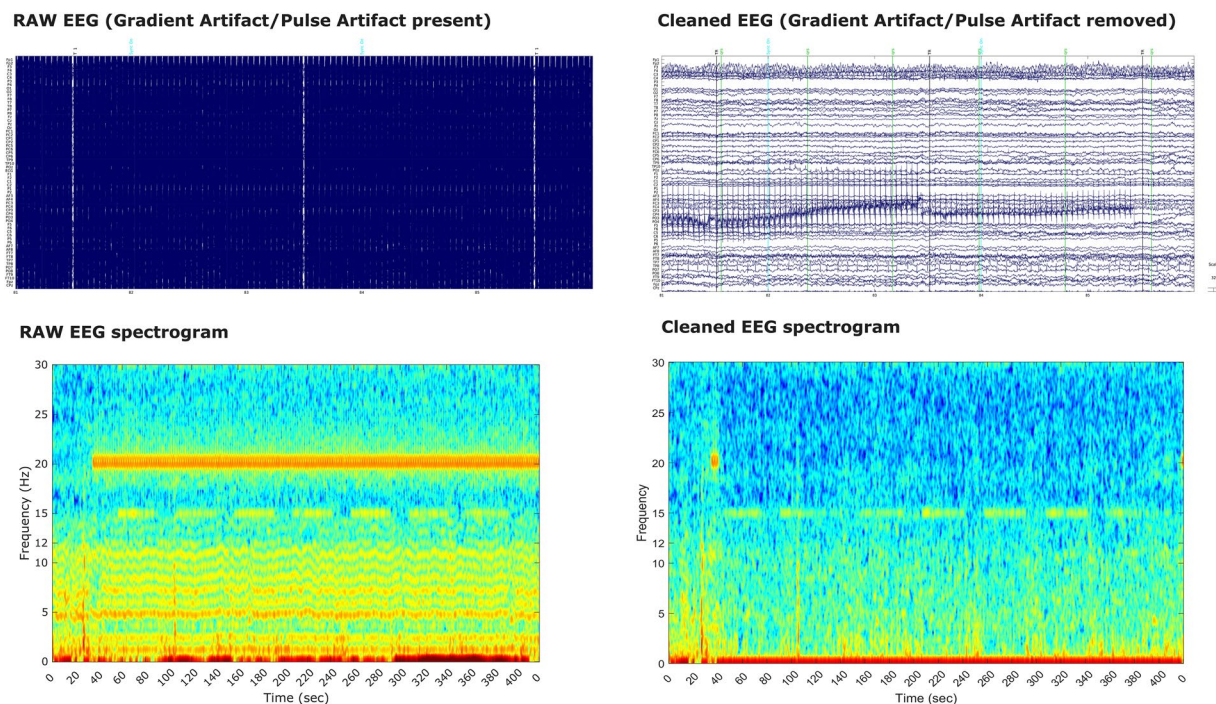


Fig. 3 Visualization of raw and cleaned EEG data (5 s) along with power spectrograms of Oz channel (400 s) of a single subject during the checkerboard task.

removal of pulse artifacts/BCG by applying template subtraction based on the median artifact; (iii) application of a bandpass filter with a cutoff frequency range of 0.5 Hz to 100 Hz through EEGLAB function.

In the Scan ON setting, the preprocessing procedure is the same as in the Scan OFF session. The additional step involves removing noise caused by gradient changes according to the fMRI pulse sequence using the FMRIB plugin. For gradient artifact removal, an averaging window size of 30 was used for artifact template creation. This value (30) is the default window size in the method. In addition, since we found that applying adaptive noise cancellation (ANC) to some files from the gradCPT task also removed parts of the data, was not applied to minimize data loss. The process includes: (i) gradient artifact removal; (ii) detection of QRS complexes/heartbeat through the ECG channel; (iii) removal of pulse artifacts/BCG; (iv) bandpass filtering with a cutoff frequency range of 0.5 Hz to 100 Hz.

Gradient artifact removal. The main noise source in simultaneous EEG-fMRI data is the gradient artifact, which is known to be 400 times larger than the lowest amplitude of raw EEG data. Like previous studies⁵, we also used the FASTER method of the FMRIB plugin to remove gradient artifacts¹⁷. This method calculates an average template based on recordings of the TR start time during data collection and subtracts this template from the raw EEG data.

Pulse artifact/BCG removal. Generally, ECG data is known to exhibit a more pronounced T wave as the field strength inside the MRI scanner increases. We used the FMRIB plugin's algorithm to detect and align these QRS waves, thereby correcting for false signals. This process involves calculating a median signal to create an artifact template, which is then removed from the raw data. Visualization of raw EEG and cleaned EEG is shown in Fig. 3.

fMRI preprocessing. The fMRI data was formatted according to the Brain Imaging Data Structure (BIDS) guidelines¹⁹ and preprocessed using the fMRIPrep pipeline (v23.0.1)²⁰. The anatomical T1w image was corrected for intensity non-uniformity, skull-stripped, segmented, and normalized to the standard Montreal Neurological Institute space (MNI152NLin2009cAsym). The runs for the fMRI data were all preprocessed using the same pipeline. Structural MRI images were defaced using pydeface²¹, (<https://github.com/poldracklab/pydeface>) to remove identifiable facial features while preserving brain anatomy. For each of the 10 functional runs, preprocessing steps included slice-timing correction, coregistration of the blood-oxygen-level-dependent (BOLD) reference image, and resampling of BOLD time series into MNI152NLin2009cAsym space. As part of nuisance regression, six motion parameters and their derivatives, global signal, framewise displacement (FD), and six aCompCor components were included as regressors. Polynomial regression up to the second order was also applied to account for any residual variance. Frames with FD exceeding 0.5 mm were identified as motion outliers. If more than 12% of the total TRs in a run were flagged as outliers, the entire run was excluded from analysis. Three probabilistic masks—CSF, WM, and a combined CSF + WM—were created in the anatomical space. Initially, voxel-wise scaling was applied to standardize the signals across runs. The BOLD signals were spatially smoothed using an 8 mm full width at half-maximum (FWHM) isotropic Gaussian kernel. After smoothing, scaling was performed again to achieve a temporal mean intensity of 100 across the whole brain for each run.

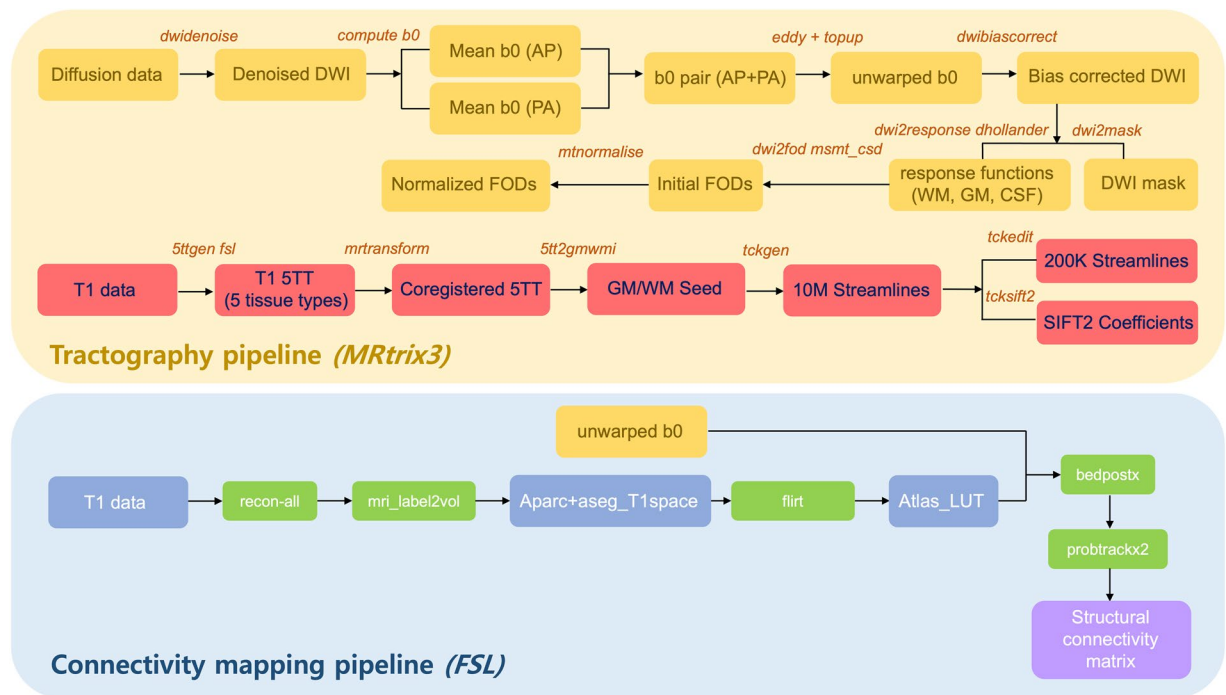


Fig. 4 Flowchart of tractography and structural connectivity mapping pipelines. The pipeline is divided into two sections. First, the tractography pipeline (yellow, using MRtrix3) processes diffusion-weighted imaging data through denoising, bias correction, and response function estimation to generate fiber orientation distributions, which are then used for whole-brain tractography and streamline filtering. Next, the connectivity mapping pipeline (blue, using FSL) uses T1-weighted anatomical data and parcellations derived from FreeSurfer to define cortical and subcortical regions. These regions are used to generate structural connectivity matrices.

Specific brain networks were analyzed by taking masks from the 7 large-scale network parcellation defined by Yeo's atlas²². The networks include visual, somatomotor (SMN), dorsal attention (DAN), ventral attention network (VAN), limbic (LIN), frontoparietal (FPN), and default mode network (DMN).

DWI preprocessing. The DWI data preprocessing was performed with MRtrix3^{23,24}. The preprocessing pipeline included denoising²⁵, distortion correction, eddy current correction, and unwarping of the data. The DWI data were denoised using the function *dwidenoise* and preprocessed using the *dwifslpreproc* function. Corrections for head motion, eddy current, and susceptibility-induced distortions were performed by applying FSL top-up and eddy²⁶. B1 field inhomogeneity corrections for DWI data were applied using *dwibiascorrect* function in the MRtrix3 (Fig. 4).

Data quality control. EEG To examine the quality of the EEG data, we used a basic quality control pipeline. For the preprocessing procedure, we used the EEGLAB package in MATLAB¹⁶. The EEG signal was processed as follows: 1) notch filtering at 60 Hz to remove power line noise, 2) bad channels were detected and removed using the *trimOutlier* function. The channel amplitude rejection thresholds were set at 10^{-4} μ V and 100 μ V. If a specific bad channel appeared more than 25% of the time across all participants (28 participants) in a task session, the common bad channel and its symmetric counterpart were excluded from all participants. After removing common bad channels, we also removed the bad channels and their symmetric counterparts for each individual participant. If the total number of bad channels removed from a participant exceeded 25% of the original total channels, that participant was excluded from the data quality assessment process.

MRI. The temporal measurements of the fMRI data included the mean FD, the root mean square of temporal changes in BOLD signal (DVARs), and the temporal signal-to-noise ratio (tSNR). The tSNR was calculated by dividing the temporal mean by temporal standard deviation over the timeseries using preprocessed (without spatial smoothing) BOLD data. We present the group-averaged tSNR brain map averaged across subjects and tasks. Across the brain, group-averaged tSNR values ranged from 0 to 168 (mean 97.5, median 100.6, SD $\approx$ 14.3), with most cortical areas showing strong tSNR levels above 80 (Fig. 5a). For correlation analysis between motion parameters, the tSNR value was computed by taking the mean across all voxels within the 3D volume.

DWI. Structural connectivity. To generate structural connectivity, we used FMRIB Diffusion Toolbox (FDT) within the FMRIB Software Library (FSL)^{27,28}. For each participant, T1-weighted anatomical images were processed with FreeSurfer to obtain cortical and subcortical segmentations and to generate a Desikan–Killiany (DK) atlas parcellation²⁹. The atlas was transformed from T1 space to each participant's DWI space using FSL's FLIRT with nearest-neighbor interpolation.

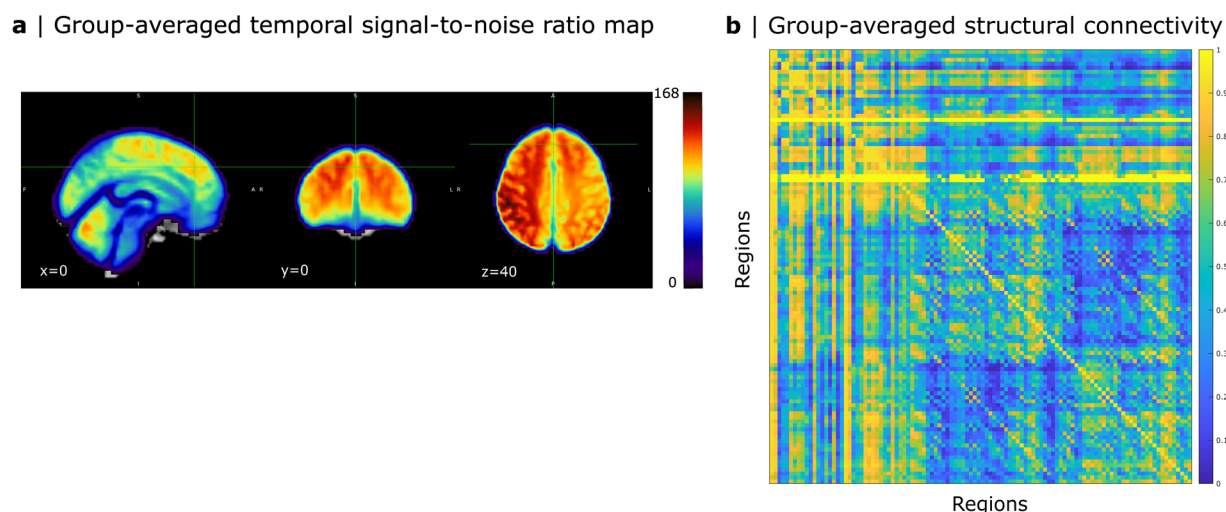


Fig. 5 Group-averaged data quality visualization for fMRI and DWI. **(a)** Group-averaged temporal signal-to-noise ratio (tSNR) maps in MNI152 space. Representative sagittal ($x=0$), coronal ($y=0$), and axial ($z=40$) slices are shown. Warmer colors indicate higher tSNR values, reflecting more reliable BOLD signal, whereas cooler colors indicate lower tSNR, commonly observed in orbitofrontal and inferior temporal regions due to susceptibility-induced dropout. Across the brain, the group-average tSNR ranged from 0 to 168 (mean = 97.5, median = 100.6, SD $\approx$ 14.3), with the majority of cortical regions exhibiting robust values exceeding 80. **(b)** Group-level 108×108 symmetric structural connectivity matrix, with each element represents the averaged probabilistic connection strength between ROI pairs.

BEDPOSTX (GPU-accelerated) was applied to estimate the diffusion parameters at each voxel³⁰. The DK atlas was then split into 108 individual ROI masks (68 cortical and 40 subcortical regions). Probabilistic tractography was performed for each ROI using PROBTRACKX2, with each ROI serving as a seed and all other ROIs as targets (Fig. 4). Connectivity values were calculated as the proportion of streamlines reaching each target ROI from a given seed. This procedure produced seed-to-target connectivity matrices (seed2all_roi*.txt) and the corresponding mean streamline lengths (seed2all_length_roi*.txt). These matrices were assembled into participant-specific 108×108 symmetric structural connectivity matrices, where each element represents the averaged probabilistic connection strength between ROI pairs, normalized for ROI size and corrected for fiber path length. Each matrix was then averaged across participants ($N=25$), resulting in a group-level structural connectivity matrix (Fig. 5b). Three participants were excluded due to insufficient ROIs ($n=2$) or failed reconstruction ($n=1$).

For each participant, two connectivity matrices were constructed: (i) a streamline count matrix and (ii) a streamline length matrix. For streamline counts, voxel-wise connectivity distributions were thresholded and averaged across all seed voxels to yield a seed-to-target connection probability. Streamline lengths were extracted from the corresponding *seed2all_length* files and averaged across streamlines. Both matrices were symmetrized by averaging the values of each ROI pair (i.e., seed $\rightarrow$ target and target $\rightarrow$ seed). The path of these metrics is 'dwi/sub-XXX_processed/connectomes/connectome_[metric_name].csv' (Fig. 6).

Statistical analysis. *EEG Analysis.* To evaluate the quality of simultaneous EEG-fMRI data, we employed a *permutation test* to compare the responses to the flickering checkerboard task with the resting state, across the Scan OFF and Scan ON sessions. The *permutation test* is non-parametric, i.e., does not rely on assumptions about data distribution, and it is resilient to noise and artifacts, making it suitable for the EEG data analysis. In this study, we focused on EEG power at 15 Hz (the frequency at which the checkerboard flickers) and compared it between the two tasks. For both Scan ON and Scan OFF sessions, the task data were randomly permuted to test whether statistically significant differences existed between conditions.

fMRI Analysis. Group-level activity map on checkerboard task was generated by applying one-sample *t*-test. For attentional task, independent samples *t*-test was used to examine trial (correct/incorrect) difference in measuring sustained attention lapse (gradCPT) and error rate differences in attentional states (gradCPT and gradCPT with imagery). For both test, normality assumption was assessed using Shapiro-Wilk test. When normality assumption was not satisfied, Mann-Whitney *U*-test was performed.

Data privacy. All personal information of the participants has been anonymized. In addition, all imaging data released through this paper has been de-identified by removing any personal identifying information.

Data Records

Data access. The dataset is available at OpenNeuro⁶. Anyone who wishes to obtain the data from this paper can download it via the site, and the guide on the website will provide instructions on how to access the data. Both raw and preprocessed data are available on the website.

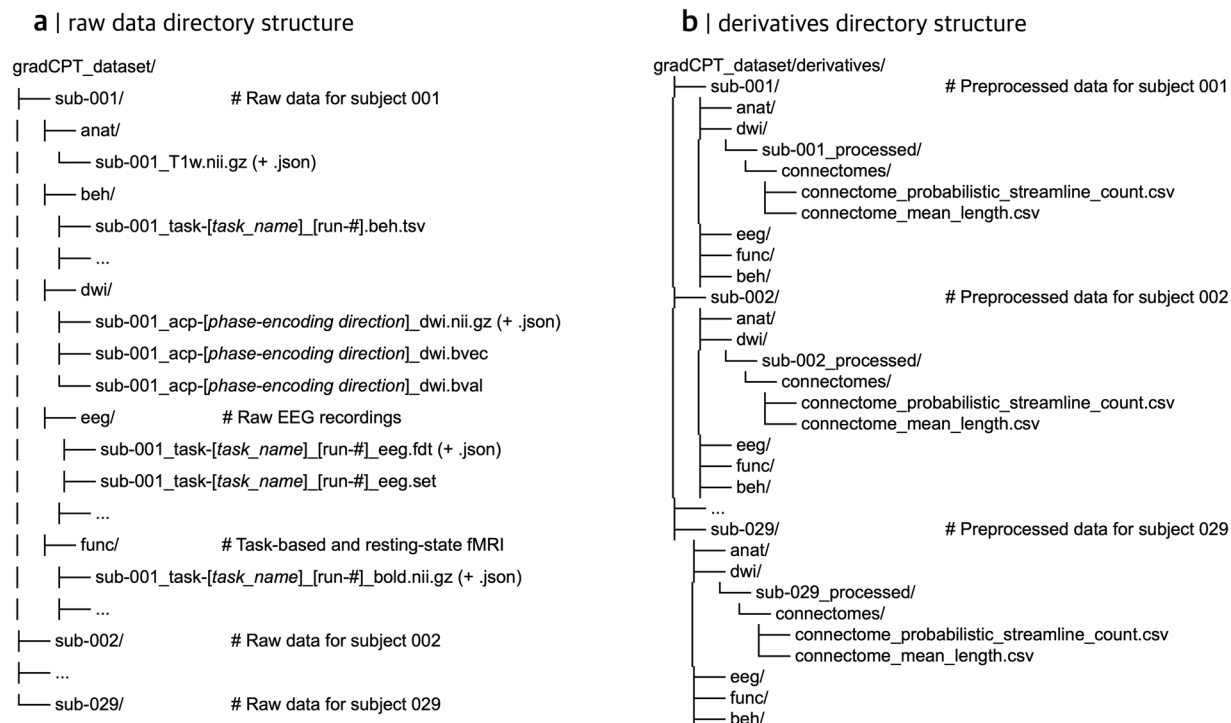


Fig. 6 Data organization and file structure of gradCPT EEG-fMRI-DTI dataset. **(a)** The root directory contains raw data for each participant (/sub-XXX), including anatomical (/anat), behavioral (/beh), diffusion-weighted (/dwi), EEG (/eeg), and functional (/func) data. **(b)** Processed data are organized under /derivatives, where each subject folder mirrors the raw data structure. Diffusion derivatives include a dwi/sub-XXX_processed/connectomes subfolder containing structural connectivity matrices (mean streamline length and probabilistic streamline counts).

Data organization. All EEG, fMRI, and DWI data were formatted according to the Brain Imaging Data Structure (BIDS)¹⁹. The resulting data structure was validated for compliance with BIDS standards using the BIDS-validator. The data can be downloaded through the OpenNeuro repository⁶, and the files are organized into a raw data folder and a preprocessed data folder (Fig. 6). Inside both the raw data and preprocessed data folders, the data are structured into five subfolders within each subject folder:

1. EEG folder: EEG data stored in EEGLAB data file format (.set).
2. fMRI folder: Contains anatomical images (anat) and functional images (func) in NIfTI format.
3. DWI folder: Stores diffusion-weighted imaging (dwi) files.
4. Behavior Data folder: Holds response information (beh) files for the gradCPT task.

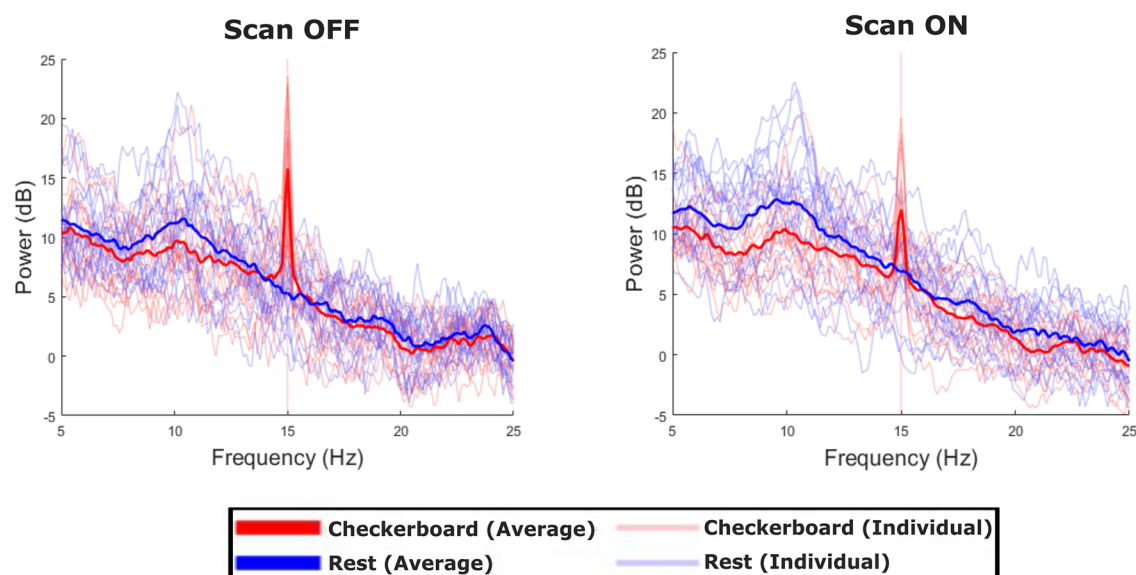
Additionally, data folder contains six sets of files (text, json, and tsv) which include dataset descriptions, survey data, such as the short form of the Big Five Inventory (BFI) questionnaire, demographic information, and readme.

Processed EEG, fMRI, and DWI data are stored in the preprocessed folder. The data structure remains the same as in the raw data folder, and file formats are unchanged. Fully preprocessed data are stored in the 'derivatives' folder, fMRI data are stored in the 'anat' and 'func' folders, while preprocessed EEG and DWI data are stored in the 'eeg' and 'dwi' folders, respectively, within the subject subfolders. Each 'dwi' folder contains a 'sub-XXX_processed' subfolder, which stores the DTI processed files and connectome results. The 'beh' folder contains the gradCPT behavioral data.

Technical Validation

EEG data validation and quality. Figure 7 presents a comparison of EEG data recorded during the checkerboard and resting state tasks under both Scan ON and OFF conditions. Figure 7a illustrates the average signal power over 35-second epochs, recorded at the Oz electrode across 19 participants, during both the checkerboard and rest tasks. Nine participants were excluded from portions of the checkerboard task due to amplitudes exceeding the range of 10^{-4} μ V and 100 μ V following gradient noise removal. Notably, the checkerboard condition exhibits prominent peaks at 15 Hz in both scanning conditions. These peaks, along with their harmonic, correspond to the driving frequency of the stimulus. This suggests that while fMRI noise may have a subtle impact on the scan-on condition, the EEG data remains valid and reliable for analysis after gradient noise removal. Additionally, when a *permutation test* was performed on the 15 Hz power (dB) for both resting state and checkerboard data in both scan conditions, significant differences between the two states were observed at a *p*-value < 0.001 level (see Fig. 7b).

a | Comparison of the power spectrum



b | 15Hz power comparison

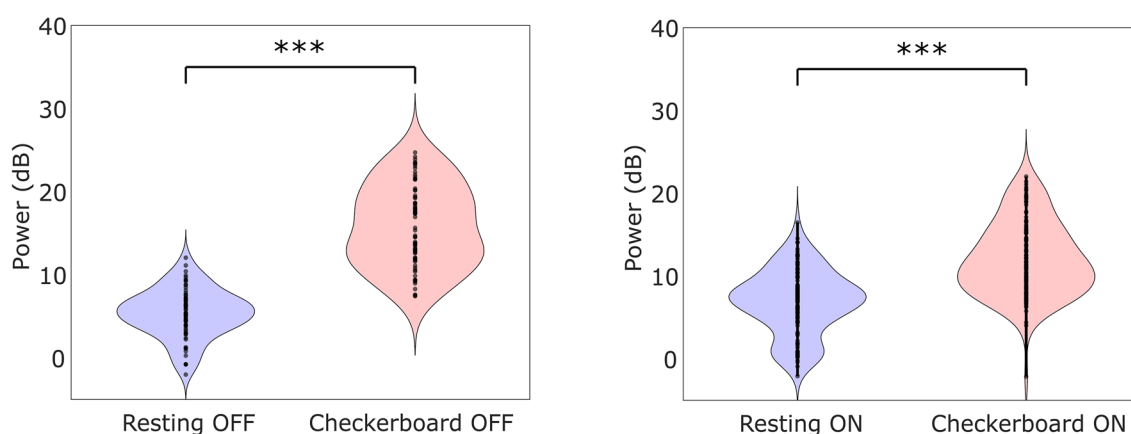


Fig. 7 EEG data validation. (a) Comparison of participants' power spectra under two scanning conditions: inside the scanner with the MRI turned off versus turned on. (b) The power at the 15 Hz frequency from EEG was compared under both Scan OFF (left) and Scan ON (right) conditions using permutation tests, which revealed a statistically significant difference between the resting state and checkerboard tasks (***) denotes p -value < 0.001).

fMRI and DWI data quality. We assessed the quality of fMRI data by calculating the average framewise displacement (FD) across 26 participants (Fig. 8a). Two participants were excluded due to missing scans. A threshold of 0.5 mm FD was applied to identify high motion TRs, and scans were considered outliers if more than 12% of their total TRs exceeded this threshold. As shown in Fig. 8a, all participants' motion remained below the 0.5 mm threshold for all scans. The histogram indicates that the mean FD primarily ranged between 0.07 and 0.11 during most task scans, while the line plot shows that the average FD remained below 0.15 throughout all tasks. For diffusion-weighted imaging (DWI), the movement parameters were quantified for each shell using Mrtrix3 and FSL³¹, focusing on root mean squared (RMS) movement values²⁶ (Fig. 8b).

fMRI data correlations. We examined correlation among several quality metrics within the fMRI modality using Spearman's ρ , including mean FD, median FD, DVARS (temporal derivative of time courses), and tSNR (temporal signal-to-noise ratio). The correlation between the motion parameters (mean FD, median FD) and tSNR/DVARS was analyzed to assess the effect of motion on image quality. Higher motion is commonly associated with a decrease in tSNR and an increase in DVARS variability, indicating instability in the BOLD signal. This analysis is important to assess data reliability by identifying whether the signal quality decreases with motion. There was a strong positive correlation between mean and median FD ($\rho = 0.92$, p -value < 0.001), while tSNR showed a negative correlation between FD measures (mean FD: $\rho = -0.53$, p -value < 0.001 ; median FD: $\rho = -0.43$,

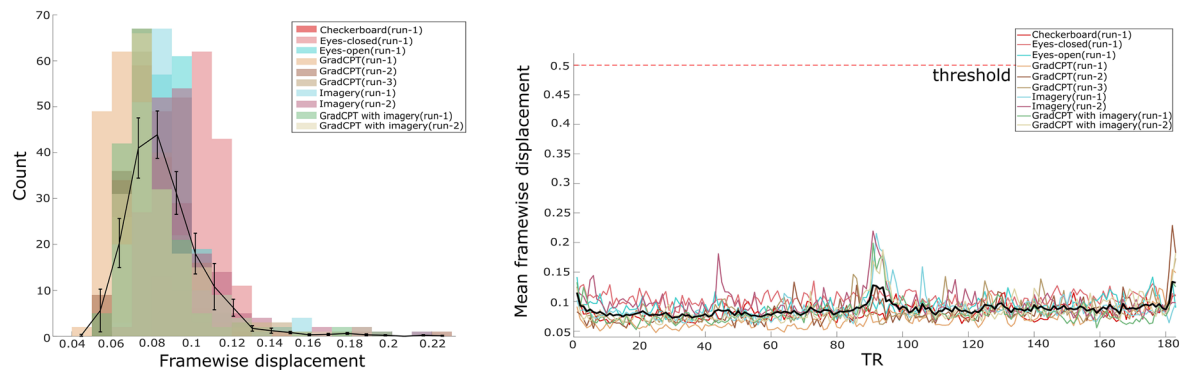
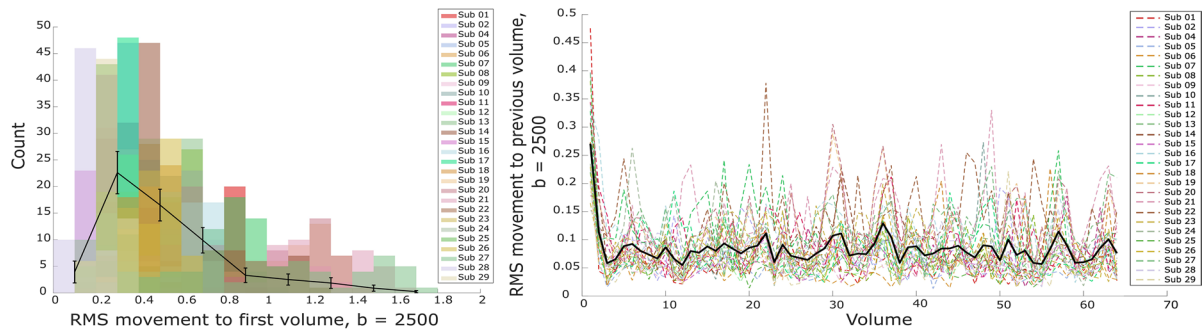
a | fMRI movement parameters**b | DWI movement parameters**

Fig. 8 Data quality for fMRI and DWI data across the subjects. For fMRI data, framewise displacement (FD) (a) was averaged across the 26 subjects per task scan and displayed in the histogram. The black solid line indicates the mean of FD count for all tasks at 0.01 interval and the error bar indicates the standard error of the mean (SEM). The line plot shows averaged FD across repetition time (TR) for 26 subjects per task. The dashed horizontal line in red indicates a threshold where a scan value > 0.5 was considered an excessive motion of participants for exclusion. The solid black line represents the averaged FD of all tasks per TR. For diffusion-weighted image data, movement parameters (b) of 28 subjects were obtained from FSL. RMS movement to the first volume indicates the histogram of root mean square (RMS) voxel-wise displacement in comparison with the first volume across all shells. The solid black line and error bars represent the average value and SEM across individuals at an interval of 0.2. RMS movement to the previous volume indicates a line plot of RMS displacement in each volume compared to the previous volume. The solid black line represents the average value across individuals per volume.

p -value < 0.001). A weak correlation was observed between DVARS and tSNR ($\rho = 0.25$, p -value < 0.001), but no correlation was observed between DVARS and mean FD ($\rho = -0.02$, p -value $= 0.736$) or median FD ($\rho = -0.04$, p -value $= 0.621$) (Fig. 9a).

We evaluated the connectivity stability using individual specificity, the degree to which an individual's connectivity profile is more similar to their own data across time than to that of others. The rationale of this approach stems from the concept of functional connectome fingerprinting^{32,33}, allowing us to measure the consistency of connectivity patterns over time. During the Checkerboard (CB) task specifically, we performed a correlation-based analysis at the subject level using the Yeo 7-network parcellation. For each subject, we first extracted the mean BOLD time series from each of the 7 Yeo networks. To assess within-subject reliability, each subject's time series was divided into two equal halves (first vs. second half of TRs). Connectivity matrices (ROI $\times$ ROI Pearson correlation) were computed separately for each half, vectorized by extracting the upper-triangle, and correlated with one another to yield a split-half correlation coefficient for each subject. To assess between-subject similarity, we computed each subject's connectivity matrix using the entire time series, vectorized it, and correlated it with the group-average connectivity matrix derived from all other subjects (leave-one-subject-out approach). This yielded a between-subject correlation coefficient for each subject. Within-subject and between-subject correlation coefficients were then compared using a paired-sample t -test across subjects ($N = 19$). The mean within-subject correlation ($M = 0.841$, $SD = 0.08$) did not significantly differ from the mean between-subject correlation ($M = 0.869$, $SD = 0.05$), $t_{18} = -1.460$, p -value $= 0.162$ (Fig. 9b).

Multimodal data integration. To test the integration of multimodal data, we used an EEG signal (Oz signal) averaged across 19 subjects during the checkerboard experiment. The Oz signal served as a regressor to map BOLD activity for each participant. We performed a whole-brain voxel-wise general linear model (GLM) analysis for each subject. The group-level one-sample t -test confirmed the average brain activity in a specific region across participants (Fig. 10a). The results show consistent activation in the occipital lobe.

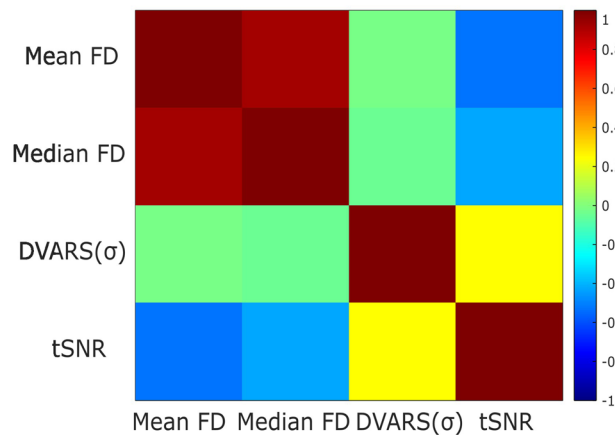
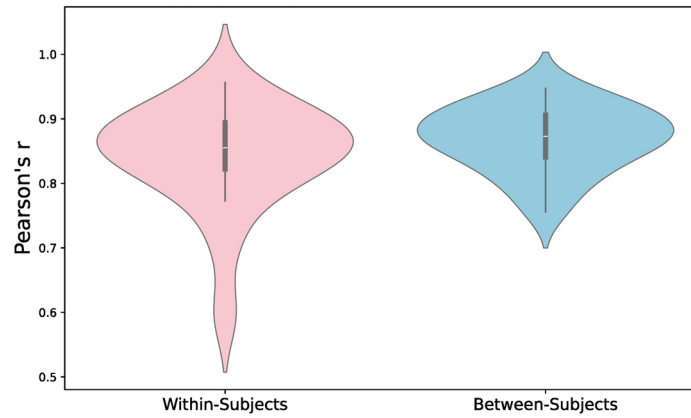
a | MRI quality metrics correlations**b | Correlations between ROIs**

Fig. 9 Correlation within MRI quality measures. **(a)** A strong positive correlation was shown between mean FD and median FD measures, while a negative correlation was shown between FD measures and tSNR. **(b)** Violin plots show the distribution of Pearson correlation coefficients from ROI-based functional connectivity, comprising within-subjects and between-subjects.

To compare with EEG-based activation map, we also employed an ideal block design regressor convolved with a gamma variate hemodynamic response function in a whole-brain voxel-wise GLM analysis (reflecting changes in cerebral blood flow). The resulting group-level activation map is shown in Fig. 10b. Both approaches produced comparable activation patterns in the occipital lobe, indicating that the model accurately captured task-related brain responses.

fMRI: Measure for sustained attention lapse. To validate the replication of gradCPT task, we examined whether the findings from the previous studies¹³ were similar in the simultaneous EEG-fMRI setting. As reported previously, we investigated whether pretarget brain activity could predict sustained attention decline (lapses) during the gradCPT by analyzing and comparing BOLD signals from correct and incorrect trials across three brain networks: the default mode network (DMN), dorsal attention network (DAN), and visual network (Fig. 11a). Out of 28 participants, 21 were included in the analysis, with 7 excluded due to poor behavioral performance (2 participants) and noise issues in EEG preprocessing (5 participants). In the case of noise issues, participants were excluded if, after removing the fMRI gradient from the EEG data, the signal amplitude was smaller than 10^{-4} μ V or larger than 100 μ V, and the amount of data that met the amplitude rejection thresholds was insufficient for analyzing the gradCPT. Correct trials included correct commissions (correctly responding to a target city) and correct omissions (correctly withholding a response to a non-target mountain). Incorrect trials included commission errors (failing to withhold a response to a mountain) and omission errors (failing to respond to a city). Although omission errors averaged 5% across participants, they were still included as incorrect trials.

Attentional states were defined based on reaction time (RT) variability during correct trials. Periods with high RT variability were categorized as “out-of-the-zone”, while those with low RT variability were classified as “in-the-zone”. Independent paired t -tests were performed for BOLD signals and behavioral measure.

The overall analysis, without separating into in-the-zone and out-of-the-zone states, revealed higher DMN activity and lower visual region activity before incorrect trials compared to correct trials. Specifically, DMN activity was significantly higher for incorrect trials ($t_{20} = -4.63$, $p = 0.0001$), while visual region activity was lower for incorrect trials ($u_{20} = 341.0$, $p = 0.0038$).

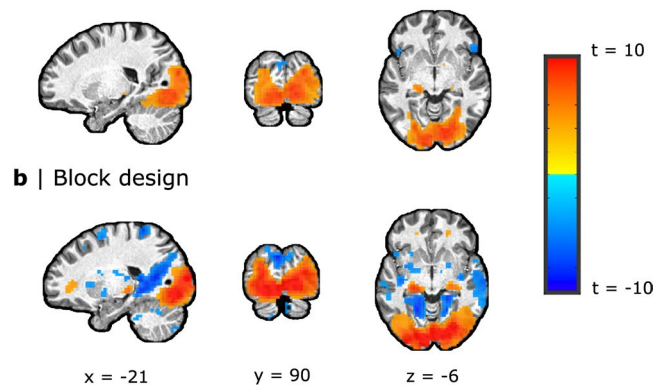
a | EEG

Fig. 10 Group-level activation map to test multimodal data integration. Group activation map from voxel-wise one-sample t -test for the checkerboard task using (a) the group average EEG Oz signal as a regressor or (b) a block design convolved with the hemodynamic response function.

When separated into in-the-zone and out-of-the-zone states, further patterns emerged. Notably, in the out-of-the-zone states, incorrect trials were preceded by higher DMN activity compared to correct trials ($t_{20} = -3.46$, $p = 0.0038$). For the visual network, correct trials exhibited higher activity than incorrect trials ($t_{20} = 2.35$, $p = 0.0355$). However, in the in-the-zone state, no significant difference was observed between correct and incorrect trials in the DMN and visual network. In fact, the effect of DMN activity before attention lapses were marginally significant. This indicates that sustained attention (attention lapses) may be predicted by higher pretarget activity in default regions and lower pretarget activity in visual regions. According to our findings, an increase in DMN activity before incorrect trials not only indicates a deviation from an optimal intermediate level as proposed in Esterman *et al.*¹³, but may also reflect further disengagement from the task-related brain networks when attention is already diminished.

To assess the robustness of these findings, we additionally examined the results without applying global signal regression (GSR). While the overall pattern remained consistent—showing increased DMN activity and decreased visual activity prior to incorrect trials—the statistical significance was weaker. In the combined analysis (zone-in + zone-out), visual activity remained significantly lower before incorrect trials ($t_{20} = 3.43$, $p = 0.0042$), and DMN activity was significantly higher ($t_{20} = 134.00$, $p = 0.0458$). In the out-of-the-zone state, visual activity was still significantly greater before correct trials ($t_{20} = 3.18$, $p = 0.0085$), but the DMN effect was not statistically significant ($t_{20} = -1.18$, $p = 0.2466$). These results indicate that the core directional effects are preserved without GSR, although the statistical strength of the DMN-related findings is enhanced when GSR is applied.

The behavioral analysis also examined error rates across the two attentional states: “in-the-zone” and “out-of-the-zone”. Considering the first run as a practice trial, we analyzed with the last two runs. Under both task conditions - gradCPT (gradCPT only) and gradCPT with imagery (dual-task) - more commission errors occurred during out-of-the-zone (34.27% and 37.54%) than during in-the-zone (28.35% and 31.25%) periods ($t_{20} = -4.20$, $p = 0.0009$; and $t_{20} = -3.79$, $p = 0.0012$). Additionally, higher omission error rates were found during out-of-the-zone (7.64%) compared to in-the-zone (5.59%) periods under the gradCPT condition only ($t_{20} = -3.23$, $p = 0.0084$) (Fig. 11b). During gradCPT with imagery, there was no significant omission error rate differences between out-of-the-zone (8.10%) and in-the-zone (7.75%) periods. These results suggest that inhibiting response becomes more difficult when participants are mind-wandering. We conducted comparative analysis between the two task conditions, but we could not find significant difference between gradCPT only and dual-task conditions.

Our results show that the decline in performance, particularly the failure to inhibit responses, occurs across both attentional states when interrupted by internal thoughts. However, the performance drop is more pronounced during out-of-the-zone periods. As expected, our results supported that poor performance in sustained attention is associated with high pretarget activity in DMN and lower pretarget activity in visual regions.

Limitations and technical challenges. Simultaneous EEG-fMRI data collection poses significant challenges, primarily due to the interaction between the magnetic fields of MRI and the EEG system. These interferences introduce various artifacts that must be addressed to ensure accurate data.

One limitation of our study is related to the imaging sequence: the simultaneous acquisition of EEG and fMRI data precluded the use of faster repetition times (TRs). Due to radiofrequency (RF) power deposition, there is a risk of heating the EEG electrodes during scanning, which could lead to burns or discomfort³⁴. To mitigate this risk, we employed a TR of 2000 ms. Additionally, as noted by Telesford *et al.*⁵, precise temporal measurements were not feasible because the delay between the stimulus computer and the projector could not be accurately quantified. However, in our study, the same stimulation presentation setting was applied to all participants to minimize the difference in delay between subjects, and the effect was considered to be limited, as the analysis mainly focused on long timescales such as BOLD responses and EEG activity.

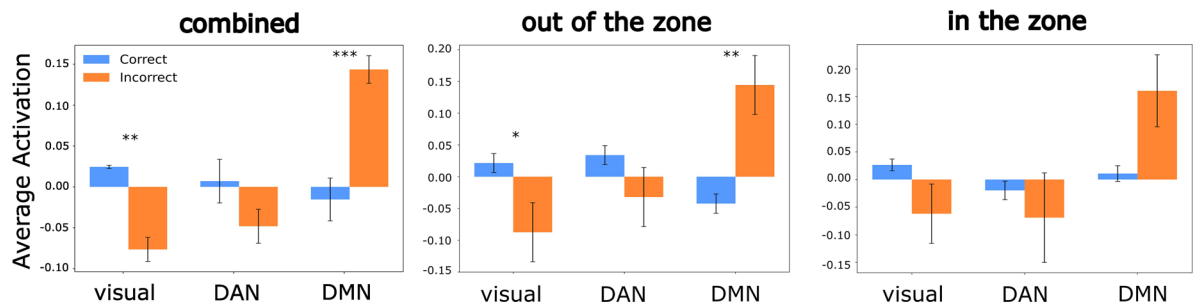
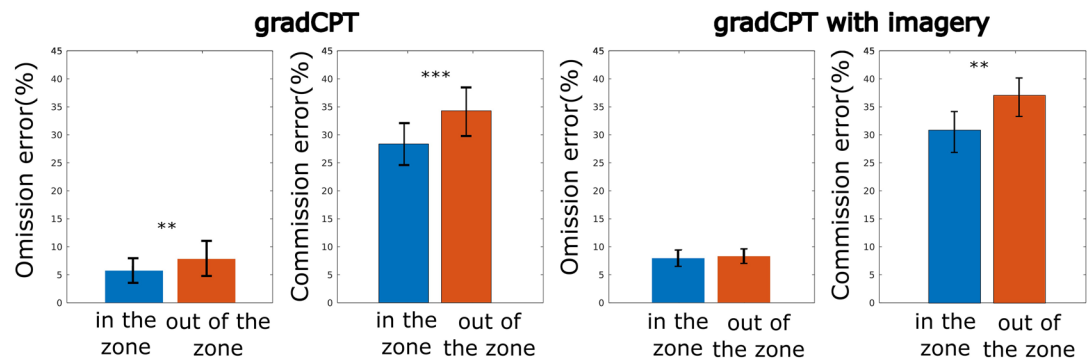
a | Predictors of sustained attention decline**b | Error rates in attentional states**

Fig. 11 Neural and behavioral correlates of sustained attention lapses. **(a)** Predictors of sustained attention decline. In the overall analysis (without separating into in-the-zone and out-of-the-zone states), incorrect trials (commission error and omission error) were preceded by higher DMN activity and lower visual region activity compared to correct trials (correct omissions and correct commissions). In the out-of-the-zone state, incorrect trials exhibited higher DMN activity and lower visual region activity compared to correct trials. **(b)** Behavioral results showing error rates across the two attentional states (in-the-zone and out-of-the-zone) under the two attentional task conditions (gradCPT and gradCPT with Imagery). Higher commission error rates were observed during out-of-the-zone periods compared to in-the-zone periods, in both gradCPT and gradCPT with Imagery conditions. Omission error rate differences were observed in the CPT condition. Error-bars represent SEM. * p -value < 0.05, ** p -value < 0.01, *** p -value < 0.001 (FDR-corrected).

Echo-planar imaging (EPI) in fMRI creates gradient artifacts by altering the magnetic field³⁵, contaminating the EEG signals. A common solution is to generate a common template, across each TR, of noises generated by the MR signals in the EEG data (usually by averaging across TRs), which is then universally subtracted from the EEG data³⁶. Additionally, if further removal of residual noise is desired, Adaptive Noise Cancellation (ANC) can be attempted. According to Allen *et al.*⁵, the ANC option has been effective in removing residual noise following gradient artifact correction. However, its effects across different types of tasks have not been thoroughly explored. For this reason, ANC was not applied to the present dataset in order to minimize data loss. Nevertheless, we confirmed that the quality of the EEG data was preserved, as demonstrated by the check-board data. Future research should further investigate the applicability and impact of ANC across a broader range of datasets.

Pulsations from scalp arteries, another source of noise, move the EEG electrodes, introducing BCG artifacts that are intensified in stronger magnetic fields³⁷. ECG signals can help detect these, but issues like signal clipping may interfere with QRS detection⁵. When the ECG channel fails, QRS detection can be done across all EEG channels, forming a median template to remove pulse artifacts. In our data, no signal clipping occurred in the ECG channel, so we performed QRS detection on the ECG channel to remove the pulse artifacts. We used FMRIB plug-in for EEGLAB as in Telesford *et al.*⁵ for MRI gradient artifact and pulse artifact removal.

The placement of EEG cables and amplifier affects EEG data quality^{5,35}, and hence should be optimized. We placed the amplifier in the inner center of the scanner bore as recommended by Brain Products^{10,35}. Additionally, excessive bending or looping of wires should be avoided. The length of the cable connecting the EEG cap and the amplifier should also be minimized to reduce artifacts⁵. In our study, two 30 cm cables were used for impedance checks, while two 10 cm cables were employed during the actual recording to minimize artifacts and ensure optimal signal quality³⁸.

The current study's limitations extend beyond the technical challenges associated with simultaneous EEG-fMRI acquisition. Notably, the sample size was modest ($n = 28$), which may limit the statistical power and robustness of the findings. In addition, participants were recruited exclusively from a university student population, which may introduce selection bias and limit the generalizability of the results to broader populations. Future studies should aim to replicate these findings in larger samples to enhance external validity. Despite

participant training and instructions, it is possible that individual differences in the degree of mental imagery participation contributed to neural response variability. Some physiological noise correction may have been limited as additional physiological signals were not recorded such as respiration or skin conductivity.

In summary, effectively mitigating artifacts arising from MRI gradients, pulsation, and, to some extent, cable placement is critical for enhancing EEG-fMRI data quality.

Data availability

The dataset generated in this study is publicly available in the OpenNeuro repository. Raw EEG, fMRI, DWI recordings, and preprocessed datasets are all accessible at <https://doi.org/10.18112/openneuro.ds006040.v1.0.1>.

Code availability

The code for preprocessing in a simultaneous EEG-fMRI setup is available on GitHub: <https://github.com/MoonBrainLab/GradCPT-Simultaneous-EEG-fMRI-DTI-Data>.

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Author contributions

Y.C., Y.L. and J.-Y.M. designed the study and experiments. Y.C., Y.L., Y.P., S.H. and J.-Y.M. set up the simultaneous EEG–fMRI and DTI experiment. Y.C., Y.L. and Y.P. were responsible for participant recruitment. Y.C., Y.L., S.M., H.K., M.C., H.S.L., Y.P. and J.-Y.M. conducted experiments and data acquisition. The processing pipeline was developed by H.K., E.J., S.H. and J.-Y.M. and data preprocessing, analysis, and quality control were carried out by Y.C., Y.L., E.J., S.M. and H.K. Data organization for upload was handled by Y.C., Y.J. and S.M. and the manuscript was written and reviewed by Y.C., Y.J., E.J., S.H. and J.-Y.M.

Competing interests

The authors declare no competing interests.

Additional information

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